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Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Compound Inequalities

< > Compound Inequalities

Toolbox

Compound Inequalities

A **compound inequality** combines two statements. You can use inverse operations to solve compound inequalities. Whatever you do to one part of the inequality, you must do to every part.

A graph of the solution to a compound inequality makes it easier to understand the answer. To make a graph, draw a short line and mark the important numbers from the solution. Then shade the part of the line that contains the solution set.

Example 1

Hide

$-6 < x + 3 < -4$

Subtract 3 from each part of the inequality.

$-6 - 3 < x + 3 - 3 < -4 - 3$

$-9 < x < -7$

The solution is $-9 < x < -7$.

The graph of the solution is:



open dot means the number is not included in the solution.

Example 2

Hide

$a + 4 \geq 10$ or $a - 1 < -5$

Solve the inequalities separately.

$a + 4 \geq 10$

$a - 1 < -5$

$a + 4 - 4 \geq 10 - 4$

$a - 1 + 1 < -5 + 1$

$a \geq 6$

$a < -4$

The solution is $a \geq 6$ or $a < -4$.

The graph of the solution is:



closed dot means 6 is included in the solution.

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Lesson 5: Inequalities

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